4 **(a)** An incoming polarised beam of monochromatic light from a light source passes through two plane polarisers. I_0 and I_1 are the intensities of the light along the central axis just before and after passing through the second polariser respectively as shown in Fig. 4.1.

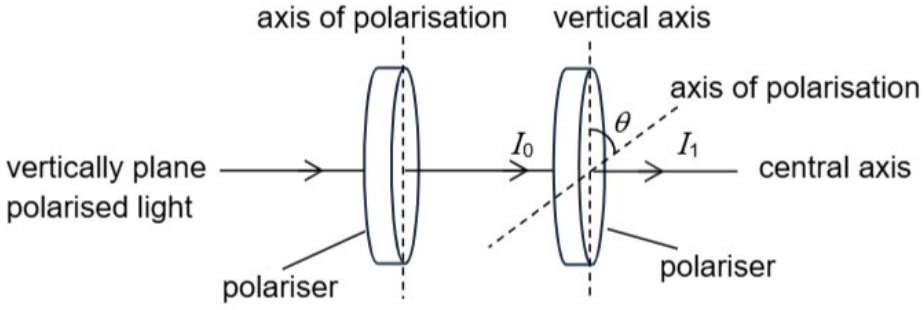


Fig. 4.1

The incoming light is polarised in the vertical plane. The two polarisers have their axis of polarisation at an angle θ to one another and the variation of I_1 with $\cos^2\theta$ is shown in Fig. 4.2.

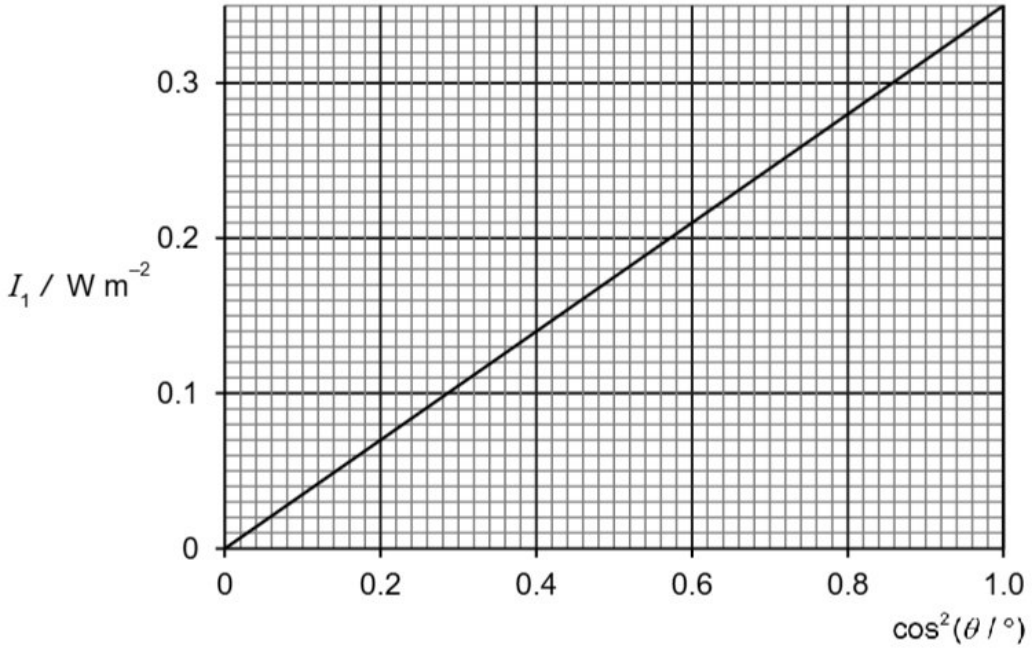


Fig. 4.2

(i) Determine two possible values of θ such that $\frac{I_1}{I_0} = 0.50$ where $0° < \theta < 360°$.

θ = ... ° and ..... ° [2]

(ii) The light source is adjusted such that the intensity of the light just before passing through the second polariser is now $\frac{1}{3}I_0$.

Draw a line in Fig. 4.2 to indicate the new variation of the intensity I_1 with $\cos^2\theta$.

[1]

(b) The polarisers are now removed and a light intensity detector is placed at different horizontal distances from the light source which is a point source of light.

The detector indicates a value of 0.20 W m^{-2} when it is at a distance of 20 cm from the light source.

On the axes of Fig. 4.3, sketch a graph to show the variation of the intensity I_d measured by the detector with distance r from the source. The graph should extend from $r = 20$ cm to $r = 60$ cm.

Include an appropriate scale for the intensity axis.

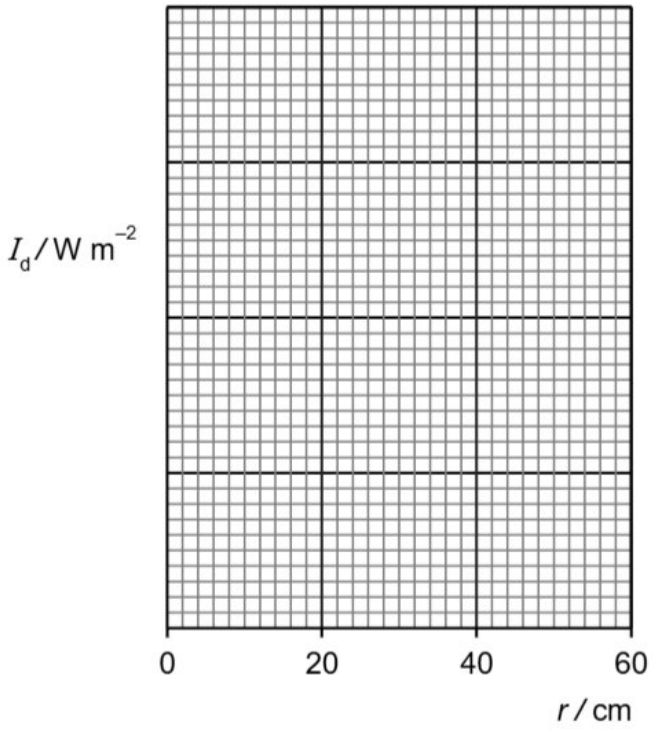


Fig 4.3

[3]

[Total: 6]